Now, both lines are intersect at P(−2,1)

(i) The line AB intersect the coordinate axis, ie) x-axis at R(−3,0) and y-axis at Q(0,3)

(ii) The line MN intersect the coordinate axis, ie) x-axis at S(−1,0) and y-axis at T(0,−1)

3.9.8 Line parallel to the coordinate axes

- If a line is parallel to the X-axis, its distance from X axis is the same and it is represented as y = c .
- If a line is parallel to the Y-axis, its distance from Y axis is the same and it is represented as x = k. (Here c and k are constants)

Example 3.43

Draw the graph of $x = 5$

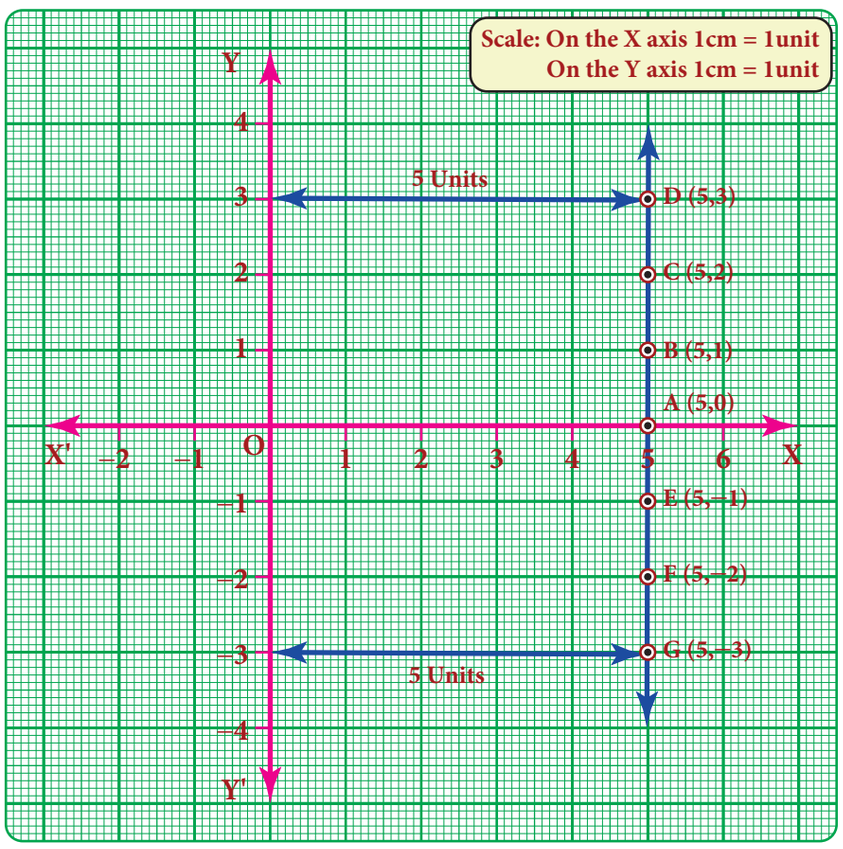
Solution:

x=5 means that x-coordinate is always 5 for whatever value of y-coordinate. So we may give any value for y-coordinate and this is tabulated as follows.

X	5	5	5	5	5
Y	−2	−1	0	2	3

x =5 is given (fixed)

Take any value for y (Why?)



Note

x =0 represents the Y axis

y =0 represents the X axis

Think

Which of the points (5, −10) (0, 5) (5, 20) lie on the straight line X = 5?

The points are (5,−2) (5,−2) (5,0) (5,2) (5,3). Plot the points in the graph and join them.We get a straight line parallel to Y axis at a distance of 5 units from the Y axis.

3.9.9 Scale in a graph

There will be situations in drawing a graph where 'y' is a big multiple of 'x' and the usual graph in units may not be enough to locate the 'y' coordinate and vice-versa. In this situation,

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